

Review

In-depth review of AI-enabled unmanned aerial vehicles: trends, vision, and challenges

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Abstract

In recent times, AI and UAV have progressed significantly in several applications. This article analyzes applications of UAV with modern green computing in various sectors. It addresses cutting-edge technologies such as green computing, generative AI, future scope, and related concerns in UAV. The research investigates the role of green computing and generative AI in combination with UAVs for navigation, object recognition and tracking, wildlife monitoring, precision agriculture, rescue operations, surveillance, and UAV communication. This study examines how modern computing technologies and UAVs are being applied in agriculture, surveillance, disaster management, and other areas. The ethics of UAV and AI applications, including safety, legal frameworks, and other issues, are thoroughly investigated. This research examines AI-based UAV applications across different disciplines, using open-source data and current advancements for future growth in this domain. This investigation will aid future researchers in their exploration of UAVs using cutting-edge computing technologies.

Keywords Aerial vehicles · Generative AI · Green computing · Traffic monitoring · Agriculture surveillance · CNN · YOLO

1 Introduction

The remote sensing technology for monitoring and photographing using unmanned aerial vehicles (UAVs) received significant acclaim for its surveillance and mapping capabilities [1]. UAVs are favored because to their affordability, user-friendliness, safety for individuals, and low training costs for operators [2]. Their advantages have resulted in a greater prevalence across diverse situations, with their superior resolution and dependable tracking features facilitating their extensive adoption. UAVs are progressively used for environmental monitoring, including the assessment of thermal sources, plant vitality, land utilization, agriculture, and air contamination [2, 3].

Mapping and surveillance operations are using drones. Their cost-effectiveness, user-friendliness, and capacity to guarantee human safety have facilitated their increasing use [4]. Their features, together with high-resolution and dependable tracking capabilities, have resulted in their growing use across several domains. UAV first used for geometric applications, have now evolved into indispensable instruments for professionals engaged in construction and data collection activities [5].

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UAVs have gained popularity for surveillance, mapping, and remote sensing [1]. UAVs are becoming more popular for many reasons, such as how cheap they are to buy, how easy they are to use, how safe they are for people, and how easy it is to train people to operate them [2]. These benefits, together with their high resolution and robust tracking features, have prompted their growing use in a variety of settings. UAVs have been used for environmental monitoring, including air pollution, land surface temperature, flood danger, forest fire, road surface distress, land terrain monitoring, pedestrian traffic monitoring, and disaster evacuation [2, 3]. For example, many individuals have an improved standard of life owing to sophisticated goods that can be controlled by mobile devices. Automobile technologies assist drivers by providing more up-to-date and precise information about traffic. A recent uptick is the use of UAVs in highway engineering for traffic monitoring and management [6].

Recent developments in computer hardware and software have made AI a crucial component in almost every engineering-related research field [7]. AI is a powerful tool for tackling difficult issues for which either no clear answers exist or traditional methods require extensive human intervention [8]. A significant difference between AI and standard cognitive algorithms is that AI can automatically extract features. This replaces expensive hand-crafted feature engineering [9].

In general, an AI job may identify anomalies, anticipate future outcomes, adapt to changing circumstances, develop an understanding of complex problems requiring enormous amounts of data, and discover patterns that a person might overlook [10]. It can use and learn from the surrounding huge data to improve UAV maneuvering. Compared with conventional optimization techniques, AI can also handle onboard resources wisely [11]. Drones that use AI typically have fully or partially automated functions. With the help of AI, drone makers can use data from devices connected to the drone to collect and use data about the surroundings and how they look [12]. AI can be used to automatically handle drones, including how they move and navigate. Several methods, including Global Positioning System (GPS) monitoring, computer vision, and machine learning algorithms, can be used to accomplish this [13]. Voice recognition, scene identification, object detection, and picture categorization are just some of the many fields where AI is making inroads, particularly when it comes to deep-learning approaches to AI [13]. Through this procedure, deep learning attempts to unearth the characteristics of unprocessed data at various levels [14]. These characteristics were then employed to represent the actual world. According to projections, the UAV market would increase from United States Dollar (USD) 26.2 billion in 2022 to USD 38.3 billion by 2027, with a 7.9% Compound annual growth rate (CAGR) over those two years. Rising acquisition of tiny drones for military applications such as Intelligence, Surveillance, and Reconnaissance (ISR) will boost small drone market development throughout the projected period [15]. Despite the growing interest in UAVs, there are still several restrictions. Many of the problems with UAVs, like high power/energy consumption and real-time needs, are also good things about edge computers and edge AI, such as low energy consumption and low delay [16]. Deep learning and unmanned aerial vehicles (UAVs) have the potential to revolutionize traffic monitoring in the transportation sector [17]. The optimal usage of AI and deep machine-learning in UAVs that are already in use and the potential future will be explored and diagnosed in this in-depth investigation, as will the large region of AI-embedded drone applications.

AI-enabled UAVs need robust computers and sensors that use a lot of energy. Energy-efficient components and algorithms are essential to green computing. A deep learning algorithm and lightweight construction can improve the performance of an AI based UAV [18]. For the purpose of energy-efficient UAV planning, it is essential to have rapid and effective control algorithms that employ data-driven strategies in real time [19]. Scientists can refine AI algorithms to lower processing demands and UAV power needs [18]. Renewable energy sources like solar panels or wind turbines may charge UAV batteries more sustainably [20]. AI is essential for UAV autonomy and navigation. The application of AI permits UAVs to navigate independently. Machine learning algorithms enable UAVs to autonomously evaluate sensor data and dynamic conditions [21]. It also aids in detecting objects filmed by UAV cameras for monitoring and rescue. AI facilitates the analysis of UAV images in a more efficient and rapid manner with regard to mapping, surveying, and inspection [22]. In multi-operational drone operations, AI keeps track of which tasks are assigned to each individual and how they can be combined [23]. In military and surveillance applications, AI is playing a critical role in detecting possible threats, identifying anomalies, and providing early warning systems [24]. Thus, these are the benefits and shortcomings of AI powered UAVs. However, conventional signal estimation is time-consuming and yields sub-optimal results. Traditional systems are adaptable and flexible to a restricted degree. Traditional signal processing methods are characterized by increased noise and restricted usability [25]. As a result, AI-enabled UAVs are required to make UAV operations more precise, rapid, and effective. AI algorithms may improve UAV flight patterns, reducing fuel consumption and emissions [23]. Researchers, industry professionals, and government organizations may collaborate to develop green computing solutions for AI-enabled UAVs. Researchers are currently working to develop eco-friendly green technology and techniques for better green UAVs [26].

A study was undertaken of Deep Learning methods based on UAV uses. They have shown the limits of the present state of UAV development. Not sufficient attention was devoted to discussing algorithms and the potential uses of UAVs in specialized industries [14]. An assessment of the ML-based UAV communication system was conducted using various algorithms. In that particular research, just few of the many potential applications of UAVs were investigated [27]. Vision-based UAV navigation system using a variety of AI technologies were examined. Several applications of computational intelligence, including search and rescue, as well as surveillance, were addressed [28]. Rescue sharing, distribution, and trajectory design uses for smart UAV base station were surveyed. Researchers analyzed potential AI-based rescue strategies [29]. Computerized vision methods based on machine learning algorithms and UAV platforms are being studied by researchers for their potential to detect and cure agricultural illnesses at an early stage. Multiple elements, including plants weeds, pests, and conditions, were taken into account in the research [30]. Researchers investigated the potential for AI-based unmanned aerial vehicle (UAV) systems applications to be used for traffic management [31], as well as monitoring [32], control [33], and detection of traffic [34]. Integrating generative AI in UAVs brings about a transformative shift in how drones' function, enhancing their intelligence and autonomy [25]. Using generative models, these UAVs can anticipate and forecast impediments, adapt their flight trajectories instantaneously, and optimize the process of making informed choices [35]. Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Conditional GANs (cGANs), and CycleGANs are recently developed methods in the area of generative artificial intelligence. These techniques are used for picture re-generation, construction, detection, and other applications. This technology enables sophisticated applications such as accurate simulation for design testing, intricate 3D mapping for surveillance, and environmental monitoring for conservation initiatives [36]. Generative AI greatly enhances UAVs' efficiency, safety, and usefulness in many industries, leading to the development of new drone applications and more intelligent and responsive aerial systems [37]. Most recent survey studies concentrated their attention on one application of AI-based UAVs. The research only provided limited examples and did not cover a significant portion of the AI and ML algorithmic landscape.

This review effort will evaluate the various uses of UAVs where AI is appropriate and examine all of the different learning algorithms that researchers are presently using. The investigation of AI-enabled UAVs is a captivating exploration of state-of-the-art technology. This detailed overview examines current trends, predicts a future in which UAVs will reshape several sectors, and discusses the issues that will be addressed. This discourse explores the trans-formative effects of AI, along with green computing and UAVs, on our society and their potential for fostering innovation and generating beneficial outcomes. This systematic review also includes a discussion of the possible data-sets with each application of UAV model with AI and the generative AI models for modern UAVs. Those are the experimental datasets and currently researchers are working on it. These datasets are applied along with Artificial Intelligence algorithms as we mentioned in every section such as CNN, YOLO and more effective algorithms. In the dataset references, image refers to a simple depiction of an item or data with visual content. Conversely, a high-quality image possesses the same attributes as a standard image, including a greater dynamic range, chroma depth, and resolution. For better accuracy, precision, recognition, and general study of a picture, a high-quality image may be the preferred choice [38]. The review process is illustrated schematically in Fig. 1, with a more comprehensive explanation of its execution provided in the Sect. 2.

The systematic literature review is set up in the manner described below: The research methodologies utilized for this investigation are discussed in Sect. 2. Types of UAV are mentioned in Sect. 3. Section 4 offers a thorough analysis and applications, while Sect. 5 lists constraints and potential future research fields. Finally, Sect. 6 discusses the summery of this study.

2 Review methodology

This section serves as an illustration of the organized and methodical approach to doing research. It includes the methods, processes, and tactics used by scholars to collect, process, and evaluate data in order to respond to certain investigations or assumptions. Figure 2 depicts the approach employed in this study. The validity, trustworthiness, and credibility of the study conclusions depend on a well-established research process.

Fig. 1 Outline of the Structured Review

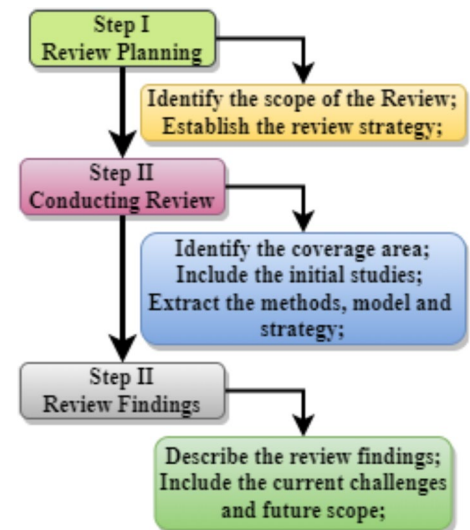
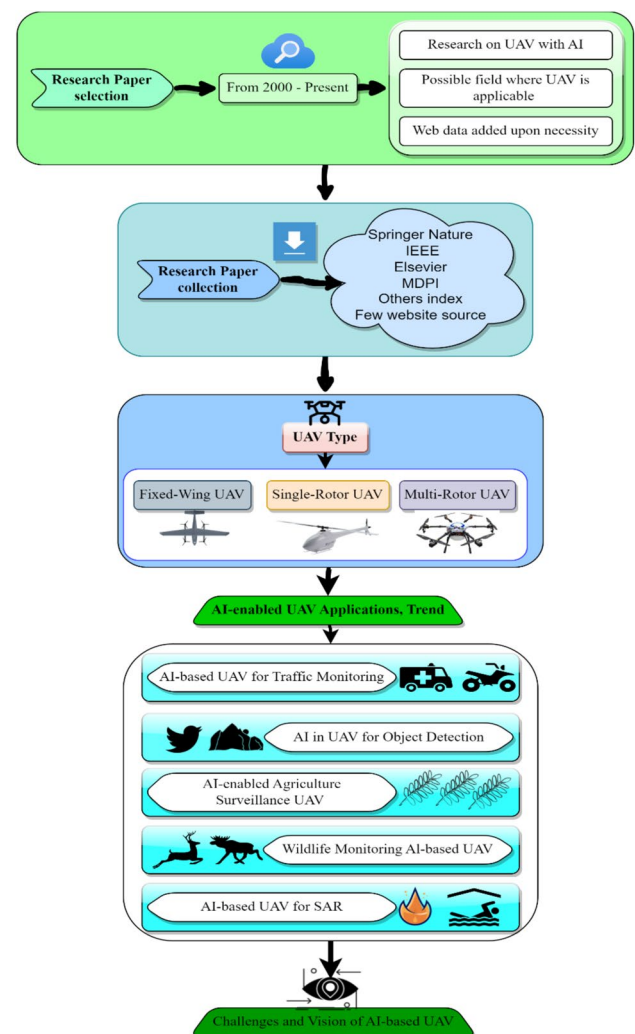


Fig. 2 AI-based UAV Review Methodology with Applications, Challenges and Limitations them ideal for aerial photography and surveillance [43]



2.1 Paper selection criteria

As with any review, we selected and included research work or paper based on a handful of criteria. These were the standards to which we adhered-

- The article should be directly related to UAV research or surveys.
- Artificial intelligence, deep learning, and UAV-related studies were also included in the selection.
- UAV-related research papers from several sub fields were also included in our evaluation owing to extensive knowledge collection.
- We picked relatively little website data when it was statistical or genuine data that were exclusively accessible on that site.

2.2 Source of literatures

A successful and instructive scientific review will comprise material gathered from publications that are reliable, well-respected, and well-organized. Consequently, the articles or information for this review study were collected from scientifically appreciated Scopus Journal indexes such as Springer Nature, IEEE, Elsevier, MDPI journals, Wiley, and many more. We acknowledge only a few articles from conferences with rigorous structural criteria. A few reliable online sources have compiled a minimal amount of statistical and up-to-date information.

2.3 Area of coverage

This study covers the period from 2000 to the present day. Several industries began implementing new applications for advanced UAVs at the beginning of 2000. After the revolution and the growth of technology, AI allows UAV to become necessary kits for practically every industry to improve surveillance, safety, security, and decision making.

3 UAV platform type

Unmanned aerial vehicles come in a wide variety, so the name "drone" is encompassing. Hexa-Copters, Quad-copters, multi-copters, and aircraft with wings all fall within this category [39]. The primary types of flying drone are as follows:

3.1 Fixed-wing UAV

A fixed-wing drone features one rigid wing that is meant to appear and function like an aircraft and provides lift instead of vertical lift rotors. Therefore, this form of drone simply requires energy to go ahead and not to maintain its airborne position [40].

Fixed-wing drones can travel farther, map much bigger areas, and stay still for a long time while keeping an eye on their target [41]. These drones have a higher ceiling and a greater payload capacity. Drones with fixed wings may be pricey. Flying fixed-wing drones typically requires training. It can be used for Aerial Mapping, Agriculture Inspection, Construction Monitoring, and numerous other applications [42].

Fig. 3 This is Fixed-Wing UAV model



Fig. 4 This is Multi-Rotor UAV model



3.2 Multi-rotor UAV

The simplest and least expensive method for keeping an “eye in the sky” is to use a multi-rotor drone. They also allow for more precise positioning and framing, making (Figs. 3, 4).

Common types of multi-rotor aircraft include tri-copters (with three rotors), quadcopters (with four), hexa-copters (with six), and octo-copters (with eight). Quad-rotors are the most prevalent multi-rotor drones. It offers superior aircraft control while in flight. Thanks to its improved maneuverability, it can go backward, forwards, sideways, and around on its axis [44]. Because of their low endurance and speed, multi-rotor drones aren’t suited for extensive aerial mapping, long-term monitoring, or long-distance infrastructure inspection like highways, pipelines, and electricity lines. They are inherently inefficient and need a lot of energy to defy gravity and maintain their airborne position [1].

3.3 Single rotor UAV

UAV with a single rotor are robust and long-lasting. They resemble helicopters in terms of construction and design. A single-rotor helicopter consists of a single rotor—similar to a large rotating wing—and a tail rotor for directional and stability control. Single-rotor helicopters are more efficient than multi-rotors, especially if they’re gas-powered. Long blades spin like wings rather than propellers, making a single-rotor helicopter efficient [45]. Single-rotor drones are costly and complicated. They tremble and are less stable or tolerant of a poor landing. Because of their technical intricacy, they need frequent maintenance [46].

4 Artificial intelligence embedded UAV

UAVs and AI are two topics that have recently attracted the interest of researchers in academia and industry [47]. Aerial drones have increased the flexibility with which operations can be carried out and activities are monitored from distant areas [48]. In addition to expanding UAV capabilities and throwing up the market to a broader variety of businesses, deploying AI and machine learning has also helped reduce the number of obstacles that must be overcome [49].

The combination of UAV with machine learning has resulted both speedy and dependable outputs [50]. Figure 5 demonstrates the present use of AI across various industries, including UAV applications. The use of UAVs in conjunction with artificial intelligence has been shown to be advantageous for real-time monitoring, the collection and processing of data, and prediction in a variety of contexts, including cities with smart, defense, farming, and mining [51].

4.1 Applications of AI in UAV for traffic monitoring

The location, speed, and direction of vehicles, as well as the number of times they traverse a particular point (like a gate, a junction, or a crossing), are just a few features that UAVs can observe and determine [52, 53]. These parameters are typically determined by the UAV placed over the coverage area. Changes in the values of the parameters discussed below can be used to identify certain occurrences [54, 55].

For instance, speeding can be determined when a vehicle’s speed is measured and exceeds a specified limit. However, traffic bottlenecks may be identified when the average speed of many cars drops below a specific limit [56]. The camera mounted on the UAV used in a surveillance system based on UAVs gathers photos of the current traffic condition using technology associated with route planning [32, 57]. The identification system, which the UAV carries, is then automatically

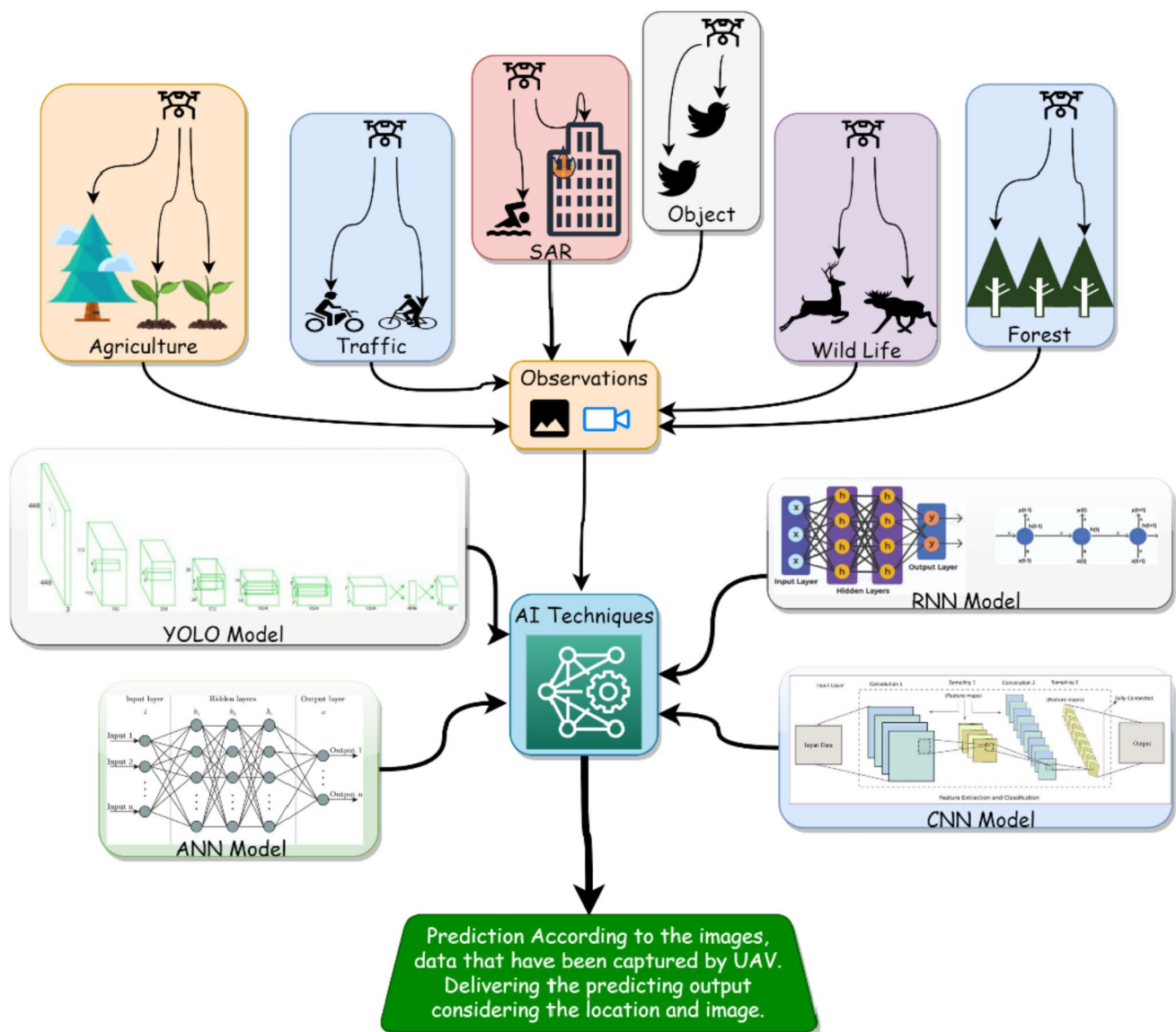
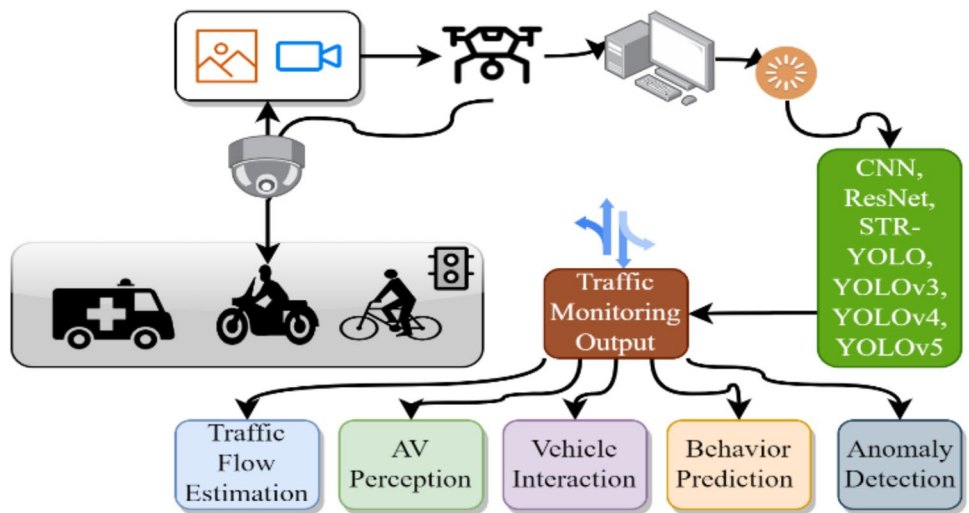


Fig. 5 AI Embedded UAV Model for real time applications with model structure

Fig. 6 Traffic Monitoring UAV model for Modern Traffic control with AI



fed these photographs. This identification system's primary skill is its capacity to assess traffic congestion [58]. The results of the recognition are sent to a traffic-management facility. The traffic managers may readily provide further analysis using those data [59, 60].

Figure 6 depicts the current AI applications for traffic monitoring and how UAVs function with it as an embedded system. The classification system that UAVs carry can be broken into two parts: one for extracting features and the other for recognizing characteristics [61, 62]. The recorded pictures are sent to the part of the system that pulls out the high-level details. The outcomes of the final recognition are then determined by the system component responsible for the recognition, and these findings are based on the extracted characteristics [63]. A reconstructed network that has convolutional layers are used as the fundamental architecture for the feature-extraction component of the system [64]. In most cases, the additional residual block cuts down on the time required for training and the level of intricacy involved [65]. Currently, several ResNet networks (e.g. ResNet-50, ResNet-101, and ResNet-110) work as a learned layer for the neural network [66]. Recently, training a 110-layer ResNet with random depth has given better results than training a 110-layer ResNet with a fixed depth, and it takes a lot less time to prepare [67]. Figure 7 shows how UAVs are used in real life to track and keep an eye on vehicles using AI methods to monitor traffic for vehicle detection, classification, and anomaly detection tasks. YOLO is a technique that offers real-time object identification via neural networks [68]. This method is used a lot because it works quickly and correctly. It has been used to monitor traffic, people, and parking meters, among other things [69, 70]. YOLO V3 switched from using Darknet-19 as the backbone network to using Darknet-53 instead [71].

Additionally, it used multi-scale estimates. Only a few academics now employ Spatial Pyramid Pooling (SPP) with YOLO V3 to identify traffic signs [72, 73]. In the system that monitors traffic, the YOLO algorithm is used for vehicle counting, detecting, and classifying, as well as monitoring traffic signs. To determine the location of a vehicle, a YOLOv4-tiny model is used [74].

Recently, an enhanced learning algorithm called TSR-YOLO has been designed to recognize the traffic sign in accordance to monitor the traffic [80]. A system capable of performing global feature extraction with a multi-branch lightweight detection head has been developed to improve the accuracy of identifying more minor traffic signs. This method is well suited to challenging weather forecasts and environments [81]. YOLOv5 (STC-YOLO) is an upgraded version of YOLO that performs better in environments with fog, snow, noise, occlusion, and blur. It is designed to monitor tiny traffic signs and vehicles [82]. The area of artificial intelligence-based UAV traffic monitoring is actively implementing green computing approaches [83]. When compared to other machine learning methods for traffic monitoring, YOLO performs better since it scans the whole scene in a single run, drastically improving speed without sacrificing accuracy [84]. When compared to traditional machine learning methods, YOLO's real-time integration of detection and classification is a major advantage. Its effectiveness and ability to generalize well across different pictures enable the completion of quick object identification tasks [85].

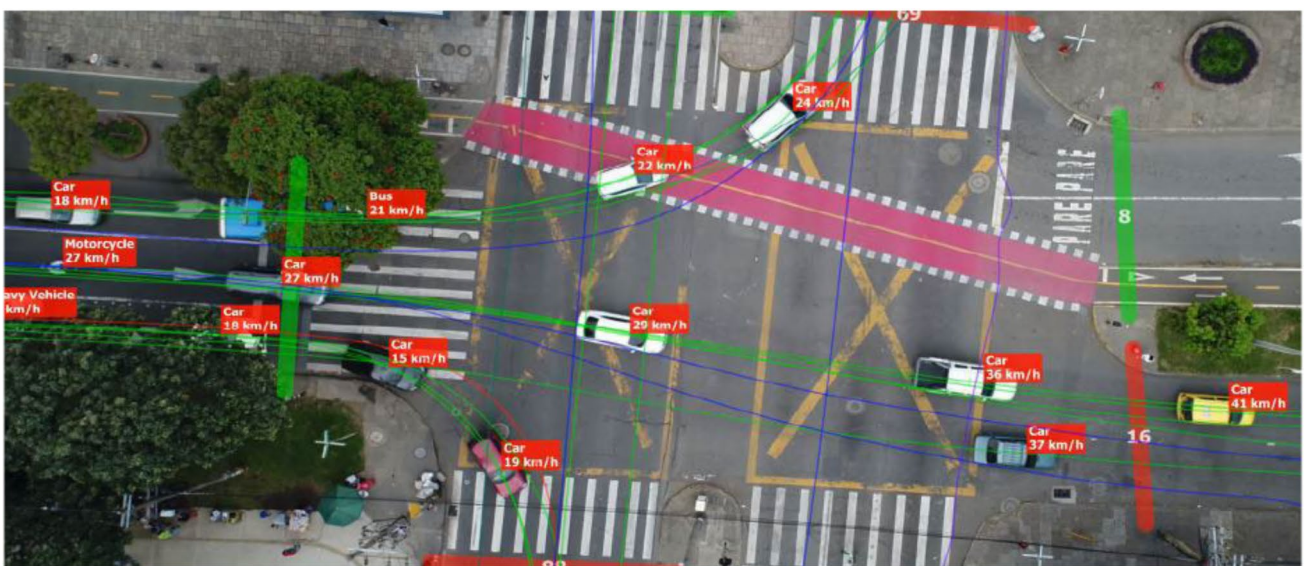


Fig. 7 Traffic Monitoring using UAV

Strategies that help reach this goal include the use of efficient hardware, the integration of renewable energy sources into ground stations, the optimization of AI algorithms to reduce the amount of computation needed, the simplification of data processing on the device itself, and the use of dynamic resource allocation. These methods aim to reduce energy consumption, decrease carbon emissions, and support environmentally sustainable aerial traffic monitoring, connecting technological advancements with environmental preservation [86]. Balancing energy usage for long-term operation is a difficulty for AI-enabled UAV. Determining the most efficient and consumptive applications of AI and ML is a formidable task. A data center located in a rural location presents a significant challenge in terms of ensuring proper ventilation. People are unaware of the secure and wholesome applications of UAVs in all industries. Combining improved natural energy with quantum computing will make green computers work better. Edge computing and decentralization can work well when there isn't a data center nearby. Ensuring the optimal utilization of green computing on UAVs requires public collaboration, awareness, and accountability.

Researchers are now employing many datasets for traffic monitoring models, and Table 1 displays the most trending datasets with plausible descriptions. All systems have certain flaws and scope for development. Despite its potential benefits, traffic monitoring using machine learning has limitations presently being studied [87]. These algorithms may provide a few wrong results when counting vehicles since they are only partially accurate [88]. The intricacy of the environment increases the likelihood that the monitoring may give faulty results under extreme weather conditions [89]. To improve the precision and accuracy of traffic monitoring, researchers are focusing on developing these [90]. Researchers are directing their attention to generative AI in UAVs for traffic monitoring. This technology revolutionizes aerial surveillance by accurately forecasting traffic patterns and congestion. It allows drones to enhance routes in real-time, enhancing traffic management and emergency response, thereby increasing the safety and efficiency of urban transportation using sophisticated, predictive analytics obtained from aerial data [35]. Several rising startups, including Shield AI, Skydio, Quantum Systems, Airobotics, and many others, are developing UAVs powered by artificial intelligence [91]. Traffic surveillance may be complicated by the presence of a great deal of video input and real-time images. The data collection method may be hampered due to urban centers. The primary obstacle is integrating the entire traffic system into a single channel. Advanced learning algorithms could be a way to handle a lot of data. Swarm Intelligence in UAVs could improve traffic monitoring efficiency. Additionally, combining Edge-Computing with a predictive traffic control system will improve the monitoring.

4.2 AI in UAV for object detection

The UAV captures the camera with its lens, and machine learning and computer vision then extract the features [92]. These algorithms are capable of detecting an item's size, form, color, and ability to recognize patterns that could locate the object specifically [93]. Researchers employ sensors such as synthetic aperture radar (SAR) and light detection and ranging (LIDAR) to collect visuals, and then artificial intelligence is used to extract information from those visuals to locate the item of interest. The SAR method allows researchers to improve the visual capabilities of unmanned aerial vehicles (UAVs) [94]. Figure 8 illustrates the existing AI applications used in the domain of object detection, specifically highlighting the integration of UAVs as an intelligent system.

Since these techniques are simple to employ in cloudy, dark, and wet settings, they are not reliant on the climate [95]. Among the portable tools for object identification is the RGB-D camera mounted to a UAV. The Parallel tracking and mapping (PTAM) method is a new development that is used with UAVs for localization and navigation [96]. This strategy is appropriate for environments and locations that are unknown [97]. One of the precise vision techniques used by UAVs to aid in object recognition is optical flow technology. It can capture images over a wide region using its long-range shooting capabilities [98]. Researchers are using numerous datasets for object detection models, and Table 2 shows the most popular with credible descriptions.

Researchers relied on template-matching techniques in the early days of UAV objective identification. The system can identify recorded objects by comparing them to a template collection containing several thousand examples [99]. For any saved or set view, the template method works very well. However, this strategy needs to catch up in terms of large amounts of data with thousands of data categories [99, 100]. At the beginning of the 2010s [101], Convolutional neural networks (CNNs) have taken over as the go-to technique for extracting the characteristics of pictures in computer vision applications [102], including picture categorization, object recognition, and semantic segmentation of photos [103, 104].

Green computing is used in AI-based UAV object identification by selecting efficient hardware, allocating resources intelligently, and processing data locally on the device to maximize the system's potential for reducing its overall energy consumption.

Table 1 UAV Traffic Monitoring Dataset

Dataset	Data type	Short description
AU-AIR [75]	Raw Video	This dataset contains video of traffic monitoring with 8 categories. GPS, time, and length are also included
Custom & Massachusetts roads dataset [76]	Images	This dataset contains images of roads, traffic, road conditions, and vehicles
Traffic Drone Data (BD) [77]	Images	This dataset contains various road conditions data such as buses, cars, and other vehicles, with annotations
Aerial Detection Dataset [78]	High-Quality Images	This dataset holds very high-quality images of buses, trucks, cars, and more. Data is split into training, test, and validation
HIT-UAV [79]	High-Quality Images	A thermal dataset containing over 40,000 frames of vehicle images

Fig. 8 Machine Learning based Object Detection UAV model for Modern Use

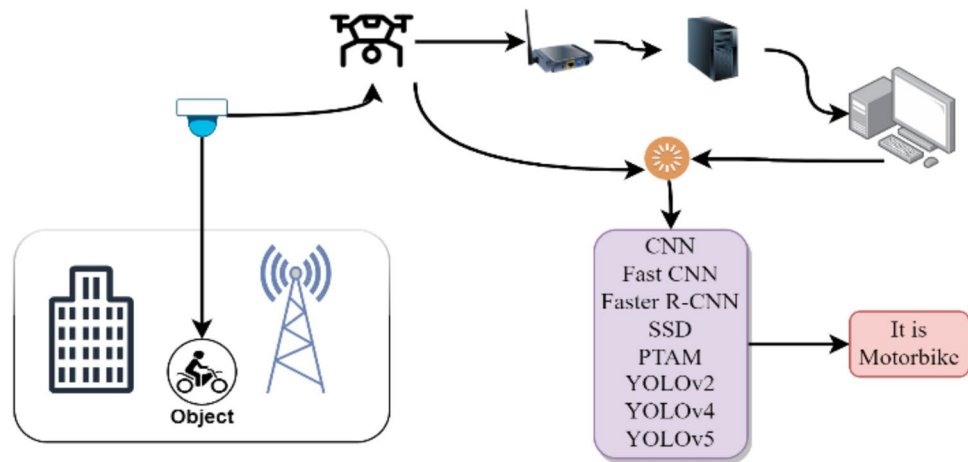


Table 2 UAV Object Detection Dataset

Dataset name	Data type	Short description
UAVAT [105]	Raw Video	This dataset contains 10 h of raw video for object detection, multiple object tracking, and single object tracking. The video is annotated
UAVOD [106]	Images	This dataset contains 10 object classes, including buildings, ships, towers, ponds, rivers, and more
Manipal [107]	Images	This dataset contains data for detecting small objects and people
Urban Zone Object Detection Dataset [108]	High-Quality Images	A combination of 3 different datasets, all well-annotated
Aerial Maritime Drone Object [109]	High-Quality Images	This dataset contains high-quality images with classes such as boats, ships, and more

This strategy lessens object detection missions' influence on the surrounding environment while preserving the capability to conduct effective surveillance. This helps to promote sustainability in aerial monitoring activities [110]. The YOLO technique was then added to UAVs for object identification and visualizing. YOLO is a one-stage technique that can quickly identify a UAV-captured picture as green computing concept. Researchers have built a YOLOv2-based ROS that can communicate with UAVs for tasks including object identification and system navigation. The method uses a simple color picture for detection [111]. To advance the object detection process to the next level, researchers have identified a few potential areas for future development. An item may be located from various vantage points and angles using a UAV. Nowadays, generative AI improves UAV object recognition by developing intricate models that accurately recognize and categorize items from aerial photos. Generative AI generates diversified training data, recreates rare situations, and optimizes detection models to enhance UAV object recognition. It creates synthetic images of animals across habitats to follow wildlife, and it models debris patterns to improve UAV-based rescue and surveillance [35]. This technology enables drones to efficiently and independently traverse intricate surroundings, observe changes, and carry out activities of objects [112]. This may affect the identification [113]. Another significant obstacle that must be overcome in the detecting process is the deformation of the moving item. Some items have intra-class variance, requiring additional training in ML [114]. The detection process can be positively effective if this variation is considered during training [115]. Scientists are striving to improve the UAV identification process using AI by addressing the problems mentioned earlier.

4.3 AI-enabled agriculture surveillance UAV system

Agriculture has far-reaching consequences for society since it is essential in maintaining human life by providing food, shelter, and employment opportunities and supplying crucial raw materials for producing a wide variety of goods [116]. UAVs have a wide variety of uses in the field of intelligent agriculture. The use of UAVs in smart agriculture allows for the

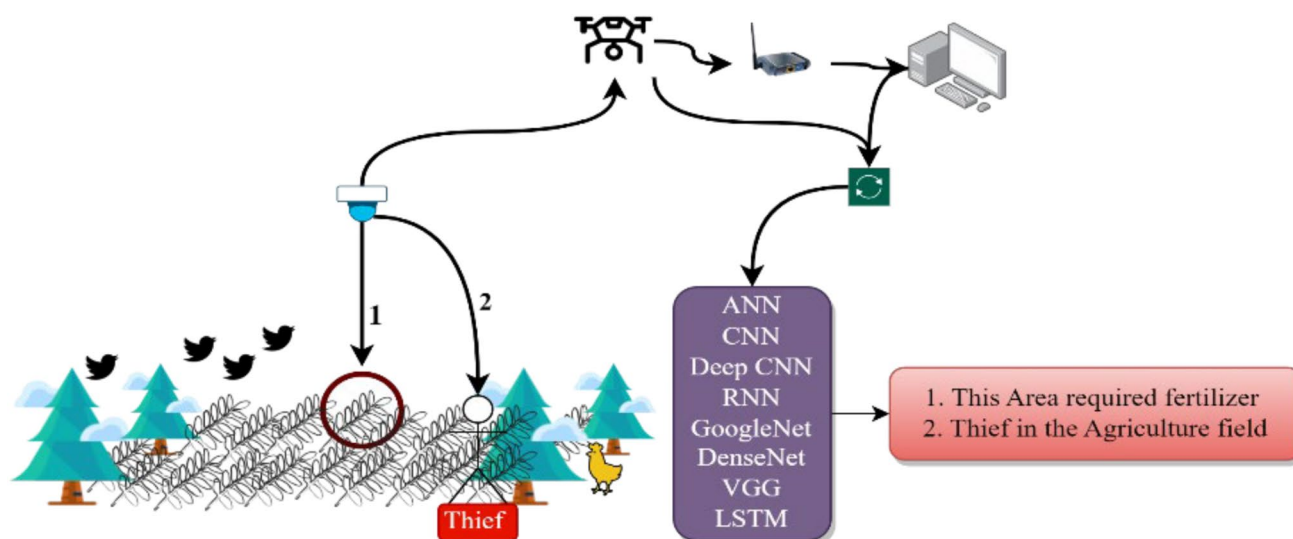


Fig. 9 Agriculture Surveillance UAV based AI model for Modern farming

Table 3 UAV agriculture surveillance dataset

Dataset name	Data type	Short description
Avo-DB [136]	Images	This dataset contains RGB images and annotated images of an avocado field
CoFly [137]	Images	This dataset consists of high-quality images of weed fields with three different classes
Paddy Field Dataset [138]	Images	This dataset contains data on various paddy conditions for low-height UAV image analysis
PlantDet [139]	Images	This dataset holds images of two different types of crops, including several leaf conditions for close UAV inspection
Crop Dataset [140]	High-Quality Images	High-resolution images of three types of crops from Kazakhstan

evaluation of agricultural field spots, the monitoring of sunlight following the growth of crops, the diagnosis of diseases, and the administration of preventive medication for plants [117, 118]. Farmers presently rely on AI-based items and applications already available for improved decision-making and earlier detection of crop conditions [119, 120]. AI enables farmers to make better decisions and more accurately assess the state of their crops [120]. The first step in intelligent farming using AI-based UAVs is harvest management, also known as yield management [121]. The authorities may also benefit from an accurate estimate of the yield since it can be used to create various strategies, including transportation needs, procurement techniques, storage facilities, and more [121, 122]. The picture data is collected from the field by UAV, then artificial intelligence technology is used to analyze it [123]. This approach helps with more accurately anticipating the yield, as well as with more intelligent watering. These AI systems provide essential data on the predicted output at an early stage [123, 124]. In the process of field analysis, ANN [125], CNN [126], and RNN [127] are often used to research the field picture to arrive at decisions and achieve precise positioning.

Figure 9 shows how AI is already being used in farming tracking and monitoring, with a focus on how UAVs are being used as part of an intelligent system. For crop surveillance models, researchers use a lot of different datasets. Table 3 shows the most common datasets and gives reliable explanations for each. One use of UAV technology in modern farming is the diagnosis of crop diseases. Deep learning algorithms, such as CNN, Deep CNN [128], GoogleNet [102], VGG [129], DenseNet [130], and many others, are currently utilized in AI-based UAV intelligent agriculture systems for detecting disease and applying organic pesticides on the spot [131].

When mapping land, UAVs are often utilized in place of survey drones. Survey drones powered by AI can produce high-resolution orthomosaics and comprehensive 3D models of regions that only have access to data of poor quality, that are outdated, or that do not have any data. They make it possible to construct high-accuracy cadastral maps rapidly and straightforwardly, even in locations that are difficult to access due to their complexity [132]. Combining GIS with AI-enhanced drone mapping opens up whole new avenues for robots to observe and comprehend the environment [133]. Superior capacities in geographical data collection, processing, and forecasting. In the process of GIS mapping,

CNN, ANN, LSTM, and Naive Bias are used extensively for image segmentation [134]. This transforms 2D photos into 3D models with high resolution for usage in GIS applications [135]. The advancement of generative AI in conjunction with UAVs transforms agriculture by enabling precision farming techniques.

It improves the monitoring of crops, forecasts agricultural yields, optimizes the use of resources, and promotes land management [37]. Producing authentic data educates models for precise disease identification and weed management, resulting in effective, environmentally friendly agricultural methods and enhanced crop yield with minimum ecological consequences [141]. Green computing is establishing a name for itself in AI-based UAV agricultural surveillance by reducing environmental effects and maximizing efficiency in resource use. Utilizing UAV hardware that is more sustainable, onboard data processing, and using renewable energy sources for ground stations makes this possible. The algorithms used in machine learning are developed to reduce the amount of required computing, which helps save energy during picture analysis and data transfer [142]. The optimal use of resources may be ensured by dynamic resource allocation and intelligent scheduling, which also reduce idle periods and overall power usage. Agricultural surveillance UAVs may improve crop management while saving power and helping sustainable farming if these green computing principles are integrated into the system [143].

4.4 Wildlife monitoring with UAV

Surveying vulnerable and invasive species to get reliable population estimates is a problematic undertaking to establish the ecological balance and sustainable growth of wildlife species [144]. Intelligent UAV systems are now used to serve forests and keep track of the animals that live there [145]. UAVs that collect geo-referenced sensor data have seen rapid adoption in the last several years, particularly for ecological surveillance and animal surveying [146, 147]. Figure 10 represents how AI is currently used in wildlife monitoring and tracking, with a focus on how UAVs are integrated into automated systems. Integrating green computing algorithms is pivotal in developing and implementing AI-based UAV wildlife monitoring systems.

The algorithms have been specifically designed to optimize eco-friendly and mitigate environmental consequences. On-board data processing and analysis capabilities are used, reducing the need for resource-intensive data transfer to ground stations [148]. By optimizing calculations and using low-power hardware components, these algorithms contribute to the conservation of energy resources in UAVs. This, in turn, leads to an extension of flight duration and a reduction in the carbon footprint associated with monitoring activities. Successful research and protection of wildlife habitats and the promotion of sustainability in aerial surveillance activities may be achieved by integrating green computing concepts into AI-based UAV wildlife monitoring [1].

Monitoring sea turtles [149], black bears [150], big land mammals (such as elephants) [151], marine mammals (such as dugongs [152]), and birds (such as flocks of snow geese [153]), as well as providing assistance for anti-poaching activities for rhinos, are all examples of ways in which UAVs may be used for governing wildlife [154, 155]. The use of several datasets is being employed by researchers for the purpose of wildlife monitoring, counting, and surveillance models.

Table 4 provides a comprehensive overview of the most often utilized datasets, accompanied by reliable and accurate descriptions. UAVs are presently used extensively for wildlife surveillance, using machine learning methods. First and

Fig. 10 Wild life Monitoring model with AI based UAV

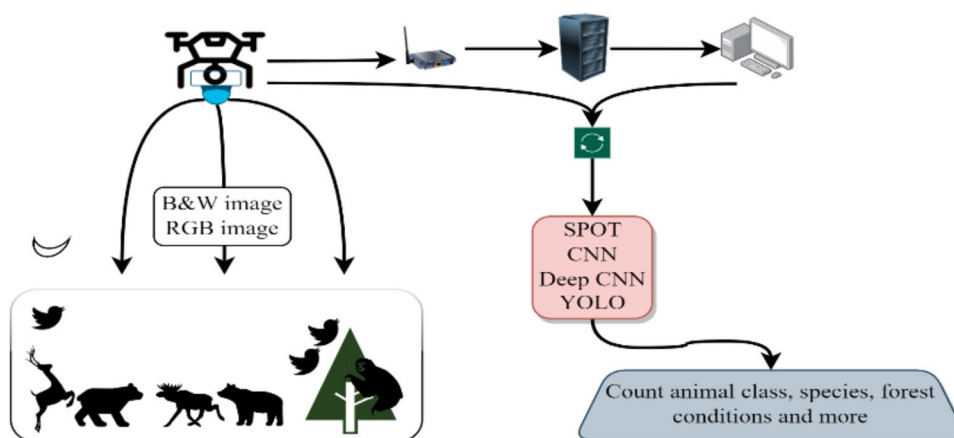


Table 4 UAV Wildlife Monitoring Dataset

Dataset Name	Data Type	Short Description
Wildlife [156]	Images	This dataset contains wild images of animals in various angles and conditions
Fast Animal Detection Dataset [157]	Images	This dataset consists of different wildlife stock and animal tracks from a wildlife reserve park
WildData [158]	Images	High-quality UAV-annotated images for wildlife detection
Drone Count Data [159]	Images	This dataset contains images of various animals for counting and analysis purposes
UAV Aided Data [160]	High-Quality Images	A wildlife surveillance dataset containing images of various animals, trees, and other species

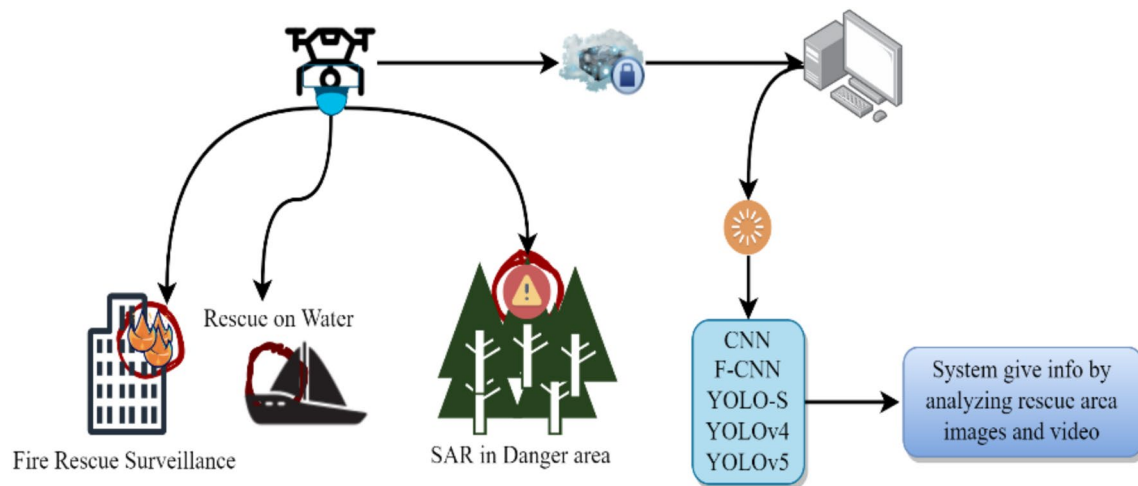


Fig. 11 SAR area monitoring using AI-enabled UAV

Table 5 Search and rescue dataset

Dataset name	Data type	Short description
SARD [179]	Images	This dataset contains annotated images of people and animals for rescue purposes
Rescue Dataset [180]	Images	Over 200 real images of search and rescue areas and locations
Dataset [181]	Images	High-quality wide UAV images for rescue operations
SAR Human Data [182]	Images	A dataset with 2,000 images of human actions for rescue operations
SAR Data [183]	High-Quality Images	High-quality thermal images for rescue operations and surveillance

foremost, UAV cameras gathered photos and video data from the forest. The picture would be in black and white, color, or RGB style for better detection and identification [161].

Regarding the data obtained from the video, the footage might be collected in its raw, night vision mode form [162]. SPOT is a program that can identify poachers in longer wavelength infrared heat UAV footage instantly [163]. SPOT searches for poachers in the simulated infrared heat picture, and identifications of poachers are represented by blue rectangles [164]. The system learns from examples to pinpoint the most important elements of photos taken in natural settings like forests and animals [165]. Currently, state-of-the-art methods, such as generative AI and UAVs, are transforming wildlife monitoring. These approaches create models that accurately identify and track animals over large areas. This technology facilitates conservation efforts by forecasting animal migrations, evaluating the condition of habitats, and identifying instances of illegal hunting, guaranteeing accurate and immediate monitoring for safeguarding and researching wildlife populations [112].

4.5 Rescue operation surveillance with UAV embedded AI

Embedded systems with machine learning architecture are now being taught to seek and detect persons, threats, or dangers inside a designated broad region utilizing UAVs [166]. CNNs of several types, including the T(Temporal)-CNN strategy [167–169], 3D-CNN [170–172], were used to identify and divide up the rescue zone. Figure 11 depicts how AI is already being employed in human and animal search and rescue, with an emphasis on the employment of UAVs as part of a smart system.

This technique is applicable for individuals in large rivers, ponds, or perilous environments. Scientists are now using a GPU-accelerated system [173], high-altitude UAV photos [174], and feature extraction based on super-pixels [175] to locate the rescue spot. The challenge of target object recognition of various sizes is addressed by the researcher's suggested model, which uses a feature pyramid network with only one stage and a densely linked set of features [176–178]. The use of several datasets is being employed by researchers in the development of models for humans, small animals,

item search, and rescue. Table 5 provides an overview of the most frequently utilized datasets in this domain, accompanied by reliable descriptions. First, UAV captures still images or moving video over a broad region, then analyzes using machine learning and deep learning algorithms [184, 185].

After the picture has been segmented, the algorithm will detect prospective locations where rescue may occur. This might occur in various environments, including a forest [186], a building, a flood [187, 188], a fire [189], and many more. Green computing methods are crucial for optimizing energy consumption and reducing environmental effects in AI based UAV rescue missions. These algorithms emphasize economy by processing data locally on the UAV, negating the need for transmission, which may be taxing on system resources [190].

In this way, the UAV may save power for crucial tasks like search and rescue, even if they are meant to run on low-power technology. UAVs with artificial intelligence are more trustworthy and environmentally friendly tools for rescue missions when they adhere to green computing principles that increase their durability. This strategy is in keeping with the larger objective of ensuring the appropriate and sustainable use of technology during times of crisis [191]. This lightweight version of the YOLO method may be found in UAVs as green computing concept, where it can handle various data classes at a high accuracy rate [192, 193]. YOLO is cutting-edge real-time software for analyzing images and videos [194], making it useful for SAR operations. A strategy that YOLO uses is analogous to the F(Fully)-CNN algorithm, which the SAR system utilizes. Researchers use YOLO-S [195], YOLOv4 [196], and YOLOv5 [197] on UAVs to carry out rescue operations. These techniques are readily available.

5 Challenges and future aspect on AI enabled UAV

Utilizing AI for UAV systems has resulted in the introduction of a multitude of innovative and resourceful solutions to an interminable array of issues [198]. Drones are utilized to gather sensitive information from hazardous environments including high winds, terrible weather, heavy rain, and multi-shaded objects [199]. A vision-based UAV navigation system using a variety of AI technologies were examined. Several applications of computational intelligence, including search and rescue, as well as surveillance, were addressed.

In addition, as these systems become increasingly autonomous and interconnected, they become potential targets for criminals looking to exploit vulnerabilities. To avoid interruptions and illegal access, it is crucial to develop solid security measures and protections. Future research in AI-based UAVs will need multidisciplinary cooperation between specialists in AI, aeronautics, ethics, law, and other fields. This cooperation is required to develop thorough regulatory frameworks that direct the use of AI-enabled UAVs while protecting against possible hazards. Research should continue to concentrate on improving AI algorithms for better navigation, judgment, and adaptability, opening the door for more advanced and trustworthy autonomous systems. Several AI algorithms are now being implemented in UAVs in order to facilitate the operation of a variety of applications. Table 6 provides an overview of the various AI algorithms currently being used in UAVs.

The use of AI with UAVs raises legal and ethical concern, mainly in human privacy, safety, and data abuse. The main concerns regarding privacy arise from the possibility of extensive monitoring, while concerns regarding security arise from the possibility of hacks and data leaks. Future regulations will need to focus on these risks and put an end to all forms of abusive, harmful, or illegal tracking so as to ensure safe and effective use of social welfare.

Table 7 lists the potential difficulties and tasks that might accelerate the development of AI-based UAVs. Also, finding new ways to use UAVs with AI, like in urban planning, environmental tracking, and crisis reaction, has the potential to change businesses and make society as a whole better. AI and UAVs working together is a growing area with much potential. Ongoing study and development will help unlock this potential and solve problems that come up along the way. As AI technology changes and new models develop, cooperation between science and technology will be critical in determining where AI-based UAVs go and how they influence our world.

6 Summary

"In-Depth Review of AI-enabled Unmanned Aerial Vehicles: Trends, Vision, and Challenges" delves into the developing scenario of AI-based UAV systems. Table 6 presents an overview of the applications of AI in several areas, with a specific focus on the current applications of UAVs. This analysis takes a look at recent developments, potential future outcomes, and the current difficulties associated with this dynamic partnership. It illustrates how AI is vital in allowing

Table 6 Review Summary of AI on UAV

Application	Algorithms (2000–2010)	Uses of AI Algorithms (2000–2010)	Algorithms (2011–2024)	Uses of AI Algorithms (2011–2024)
AI-based UAV for Traffic Monitoring	CNN, RNN [64]	Feature extraction; Real-time tracking; Traffic counting & surveillance	ResNet-50 [65], ResNet-101 [66], ResNet-110 [67], TSR-YOLO [80], YOLOv3 [71], YOLOv4 [74], YOLOv5 [82], DarkNet-19 [72], DarkNet-53 [73]	Identification of traffic, people, and locations; Measure parking space; Vehicle counting, monitoring, detection; Traffic sign detection in tough weather
AI in UAV for Object Detection	CNN [102], Fast CNN [102], Fast R-CNN [103], ANN [104]	Object detection; Color, shape, size detection; Small object detection in difficult weather	SSD [98], PTAM [96], YOLOv2 [111], YOLOv4 [113], YOLOv5 [114]	Detect small objects; Identify moving and shape-changing objects; Detection of small, multi-color, movable objects in hazy weather
Agriculture Surveillance UAV with AI	ANN, CNN [126], Deep CNN [128], RNN [131]	Crop monitoring; Crop growth measurement; Farming tracking	GoogleNet [102], DenseNet [130], VGG, LSTM [129]	Crop disease detection; Early-stage leaf disorder prediction; Thief and animal monitoring for crops
Wildlife Monitoring AI-based UAV	CNN [146], Deep CNN [147], ANN [161], RNN [162]	Animal tracking; Wild growth monitoring; Animal count & survey	SPOT [164], YOLO [163]	Animal life monitoring; Moving animal counting; Small animal species count and measurement; Hunter detection
Rescue Operation Surveillance AI-based UAV	CNN [167], F-CNN [168], T-CNN [169], 3D-CNN [170]	Detection of person, animal; Identifying danger, threat; Measure rescue condition	YOLOv5 [195], YOLOv4 [196], YOLOv5 [197]	Identify small animals and people accurately; SAR area monitoring; Assist firefighters with UAV robots; Applicable in forest, building, and field SAR operations

Table 7 Challenges and Vision on AI-enabled UAV

Domain	Challenges	Future Scope
Traffic Monitoring	- Large video feeds and real-time image processing - Data collection in urban environments - Detecting anomalies - Integrating full traffic systems	- Advanced AI for processing large data - Autonomous UAV fleets - Swarm Intelligence in UAVs - Edge-Computing integration - Predictive traffic management systems
Object Detection	- Detecting small and moving objects - Non-annotated data complexities - Diverse environment adaptation	- Developing Edge AI and Onboard Processing - Transfer Learning specific to domain - Privacy-aware object detection - Generalization and transfer ability
Agriculture Surveillance	- Data management and storage - Crop and pest identification complexities - Interoperability with farming systems - Battery life and endurance for wide areas	- Multi-spectral and hyperspectral imaging - Automated AI-based agriculture data analysis - IoT integration - Autonomous navigation and dynamic planning
Wildlife Monitoring	- Avoiding disturbance to wildlife - Difficult natural environment adaptation - Animal movement in complex weather - High-resolution data collection and storage	- Swarm intelligence - Remote sensing technology advancements - Improved conservation decision support - AI for species identification
Rescue Operation Surveillance	- Real-time situation awareness - Navigating challenging environments - Payload and endurance - Medical support via AI	- AI for disaster prediction - Robust design and redundancy - AI-assisted emergency and medical response

autonomous UAV capabilities, from navigation to object identification, and addresses applications such as wildlife monitoring, precision agriculture, rescue operations, and more.

In addition, the study discusses efficient computing strategies with energy, legal issues, ethical problems, and safety precautions. This in-depth assessment may be a helpful resource for grasping the complicated nature and transformational possibilities of AI-enabled UAV technology.

7 Conclusion

The exhaustive analysis of AI-enabled UAVs reveals the dynamic nature of this technology's landscape. The future of UAV-based applications may lie in green computing and generative AI. The study examines emerging trends that highlight the incorporation of AI, propelling UAVs to new heights of autonomy, efficiency, and applicability across multiple industries. The review shows a picture of a future where AI-powered UAVs change businesses like tracking, detecting, espionage, transportation, and emergency management. However, the path towards this vision is fraught with obstacles, such as regulatory obstacles, ethical considerations, and technical complexities. As AI and UAV technologies continue to advance in tandem, resolving these obstacles will be essential to realizing the full potential of AI-enabled UAVs for the benefit of society. AI-enabled UAVs face privacy, safety, and data usage vulnerabilities that need guidelines to avoid intrusive monitoring, hacking, and threatening tracking for social good.

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